



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VRV AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

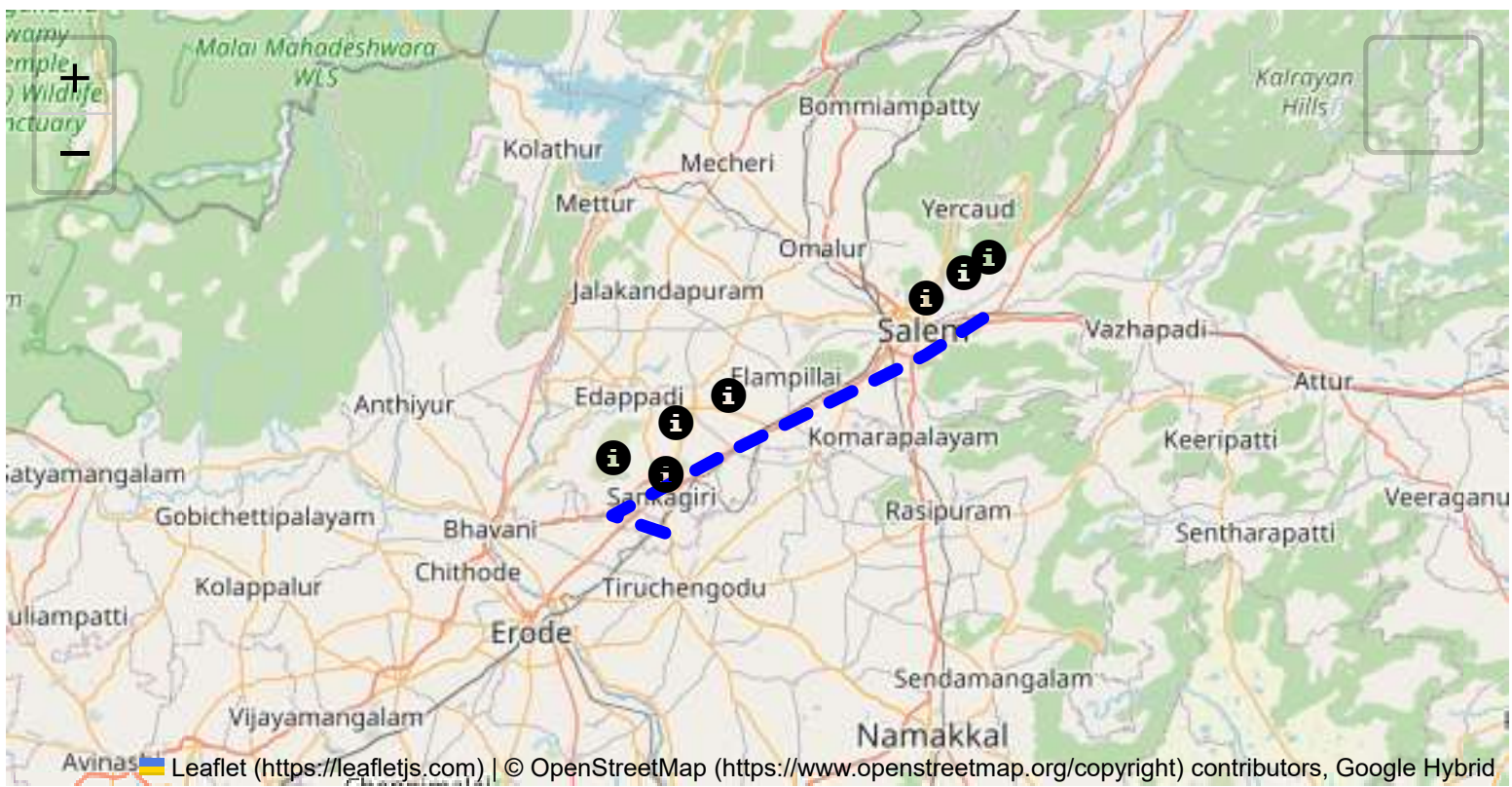
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

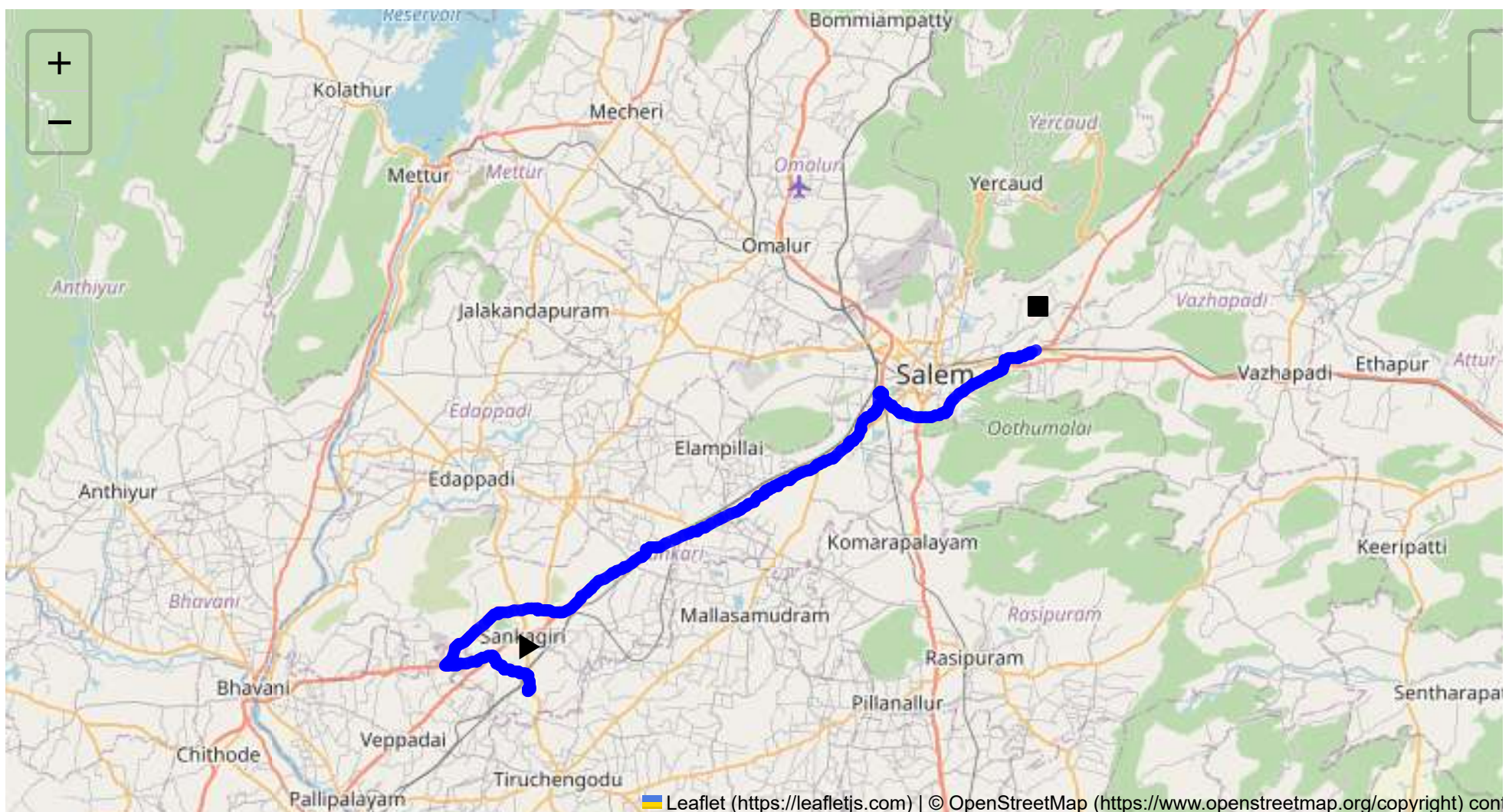
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 64.44 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.670757, 78.227951)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Masinaickenpatti passes through several key waypoints: Puthur, Padaiveedu, the Salem-Kochi Highway, Vaikundam, K R Nagar, Seelanaickenpatti, and finally Salem. The journey covers approximately 64.44 kilometers and predominantly involves highways and main roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu usually experiences a tropical climate. Travelers on this route should be prepared for:

- **Monsoon Season (June to September):** Expect heavy rains, which can lead to waterlogging and reduced visibility.
- **Summer (March to May):** Temperatures can soar, possibly leading to heat-related vehicle issues.
- **Cyclones:** Infrequent, but when they occur, can cause severe disruptions.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning from 8 AM to 10 AM and evening from 5 PM to 8 PM are prone to traffic congestion, especially around Salem and key junctions like Sangagiri and Seelanaickenpatti.
- **Congestion-Prone Areas:** Near Salem's city limits and major intersections along the Bhavani Main Road.

4. Assessment of Road Quality and Infrastructure

- **General Condition:** Mostly well-paved highways with occasional sections requiring maintenance, mainly around rural areas.
- **Infrastructure Issues:** Inadequate signage in certain rural stretches and potential for poor visibility due to narrow roads.

5. Suggestions for Alternative Routes for Emergencies

- **Via NH 544 and NH 79:** Provides a faster bypass with potentially less congestion, although longer by distance.
- **Local Roads:** Use local bypass roads around Salem for urban congestion during peak times.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Ensure valid permits for transporting hazardous materials.
- **Restricted Areas:** Avoid school zones and densely populated urban areas during peak hours.
- **Safety Regulations:** Adhere to guidelines regarding speed limits, vehicle signage, and safety equipment.

7. Overview of Historical Incidents

- **Accidents Involving Heavy Vehicles:** Have been sporadic and often due to driver error or road conditions.
- **Hazardous Material Spills:** Few recorded, mainly involving minor leaks with resolved clean-ups.

8. Environmental Considerations and Sensitive Areas

- Protected Areas:** Avoid wildlife sanctuaries and agricultural fields especially during planting/harvest seasons.
- Water Bodies:** Exercise caution near rivers and reservoirs during monsoons.

9. Analysis of Communication Coverage

- Good Coverage:** Most parts of Salem and along Salem-Kochi Highway.
- Potential Dead Zones:** Rural stretches near Vaikundam and Puthur might have weaker signals.

10. Estimated Emergency Response Times

- Urban Areas (Salem and Sangagiri):** 20-30 minutes.
- Rural Areas (Near Vaikundam and Padaiveedu):** 45-60 minutes due to the distance from emergency services.

11. Overall Summary of Risk Assessment

- Low to Moderate Risk:** Provided all regulatory guidelines are followed and awareness of weather conditions is maintained.
- Roads:** Generally good infrastructure with some rural road maintenance needs.
- Traffic:** Manageable with planning, especially during peak hours.
- Emergency Preparedness:** Contingency plans should be in place, especially for weather-related disruptions.

This assessment emphasizes the importance of complying with local transport regulations, planning for weather changes, and remaining vigilant towards traffic patterns and road conditions. Emergency response times can vary significantly, so having a robust plan is critical for dealing with potential incidents effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.45847, 77.85140	30 KM/Hr
8	Turn	Medium	11.46048, 77.85036	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr
10	Turn	Medium	11.46314, 77.84928	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45631, 77.81668	15 KM/Hr
13	Turn	High	11.49409, 77.88004	15 KM/Hr
14	Turn	Medium	11.49419, 77.88015	30 KM/Hr
15	Turn	Medium	11.49365, 77.88596	30 KM/Hr
16	Turn	High	11.49236, 77.89552	15 KM/Hr
17	Turn	High	11.49225, 77.89560	15 KM/Hr
18	Turn	Medium	11.60942, 78.10056	30 KM/Hr
19	Turn	Medium	11.60965, 78.10070	30 KM/Hr
20	Turn	Medium	11.61736, 78.10588	30 KM/Hr
21	Turn	Medium	11.61745, 78.10601	30 KM/Hr
22	Turn	Medium	11.64158, 78.11808	30 KM/Hr
23	Turn	Medium	11.64210, 78.11860	30 KM/Hr
24	Turn	Medium	11.64200, 78.12120	30 KM/Hr
25	Turn	Medium	11.64172, 78.12154	30 KM/Hr
26	Turn	Medium	11.64128, 78.12168	30 KM/Hr
27	Turn	Medium	11.64098, 78.12163	30 KM/Hr
28	Turn	Medium	11.63972, 78.12083	30 KM/Hr
29	Turn	Medium	11.62826, 78.13337	30 KM/Hr
30	Turn	Medium	11.65629, 78.20217	30 KM/Hr
31	Turn	Medium	11.66372, 78.20649	30 KM/Hr
32	Turn	Medium	11.66387, 78.20665	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
15	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
16	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
17	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
18	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
19	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
20	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
21	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
22	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
23	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
24	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
28	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
26	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
27	school	Holy Cross Matriculation Higher Secondary School	11.6635657, 78.203532	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49365, 77.88596



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49236, 77.89552



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60942, 78.10056



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60965, 78.10070



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61745, 78.10601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64158, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64210, 78.11860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64200, 78.12120



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.12154



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64128, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64098, 78.12163



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63972, 78.12083



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62826, 78.13337



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65629, 78.20217



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66372, 78.20649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66387, 78.20665
